



Lab 10

Active Filters

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10.1 Objective

To design and build Active Low Pass & Active High Pass Filters and verify with an appropriate implementation.

Note: The experiment will be done and submitted within the Lab time and there will be viva taken in this Lab.

10.2 Introduction

In previous lab you have learned to construct and analyze the behavior of passive filters, in this lab you will learn the use and application of Active filters. As its name implies, active filters contain active components such as an operational amplifier (op-amp), transistor or FETs as their main amplifying device along with some resistors and capacitors to provide a filter like performance. The question is why have we switched to active filters when resistor and capacitor-based filters were doing the desired job? We have an issue with the output of a passive filter; the amplitude of the output signal is less than that of the input signal. With passive filter circuits containing multiple stages, this loss in signal amplitude called “Attenuation” can become quite severe. One way of restoring or controlling this loss of signal is by using amplification through the use of Active Filters. These filters draw their power from an external power source and use it to boost or amplify the output signal.

10.3 Active Low Pass Filter

The principle of operation active low pass filter and frequency response is exactly the same as that for the passive filter, the only difference this time is that it uses an op-amp for amplification and gain control. These filters are used in speakers for amplification. High pass filter is used to remove unwanted sounds near to the lower end of the audible range.

The simplest form of a low pass active filter is to connect an inverting or non-inverting amplifier, to the basic circuit for an active low pass filter is shown below:

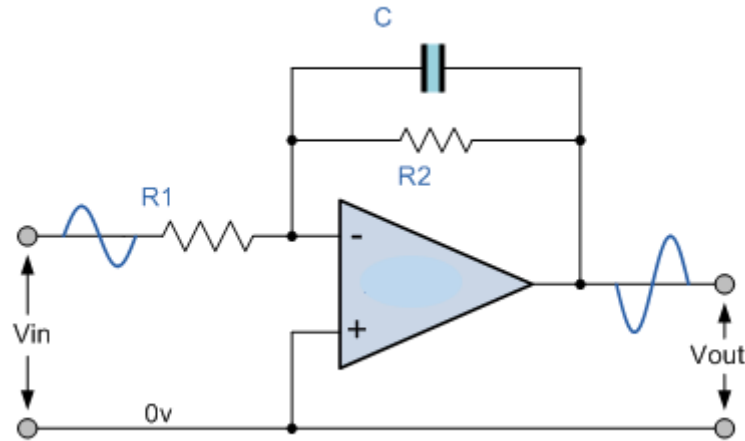


Figure 10.1: Active Low Pass Filter

10.3.1 Cut-off Frequency

If it is desired to limit certain frequencies to pass through, the same circuit could be used along with an additional component, a capacitor. The circuit in Figure 10.1 exemplify an Active Low Pass Filter. Whose cut-off or corner frequency can be calculated using equation below:

$$f_c = \frac{1}{2\pi R_2 C}$$

10.3.2 Active Filter Gain

The circuit in Figure 10.1 illustrates active filter circuitry with inverting configuration. Where resistors R_2 and R_1 set the amplifier gain¹ given as:

$$DC \text{ Gain } (A_F) = -\frac{R_2}{R_1}$$

If we take the effect of capacitor into account the gain to First-order low pass filter is given by:

$$\text{Voltage Gain } (A_V) = \frac{V_{out}}{V_{in}} = \frac{A_F}{\sqrt{1 + \left(\frac{f}{f_c}\right)^2}}$$

where:

f = the frequency of the input signal in Hertz, (Hz)

f_c = the cut-off frequency in Hertz, (Hz)

¹ For the sake of simplicity the effect of capacitor has not been taken into account.

Thus, the operation of a low pass active filter can be verified from the frequency gain equation above as:

1. At very low frequencies, $f < f_c$, $\frac{V_{out}}{V_{in}} \cong A_F$
2. At the cut-off frequency, $f = f_c$, $\frac{V_{out}}{V_{in}} \cong \frac{A_F}{\sqrt{2}}$
3. At very high frequencies, $f > f_c$, $\frac{V_{out}}{V_{in}} < A_F$

10.3.3 Frequency Response of Low Pass Filter

The frequency response of the circuit will be the same as that for the passive RC filter, except that the amplitude of the output is increased by the pass band gain, A_F of the amplifier.

The maximum frequency response of an active filter is limited to the Gain/Bandwidth product (or open loop gain) of the operational amplifier being used. Active filters are generally much easier to design than passive filters, they produce good performance characteristics, very good accuracy with a steep roll-off and low noise when used with a good circuit design.

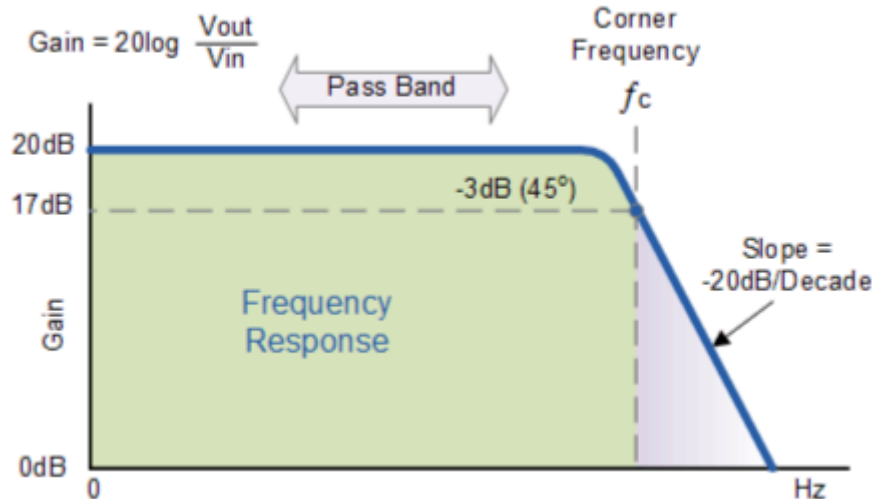


Figure 10.2: Frequency Response of Low Pass Filter

Task 1: To design an active Low Pass Filter

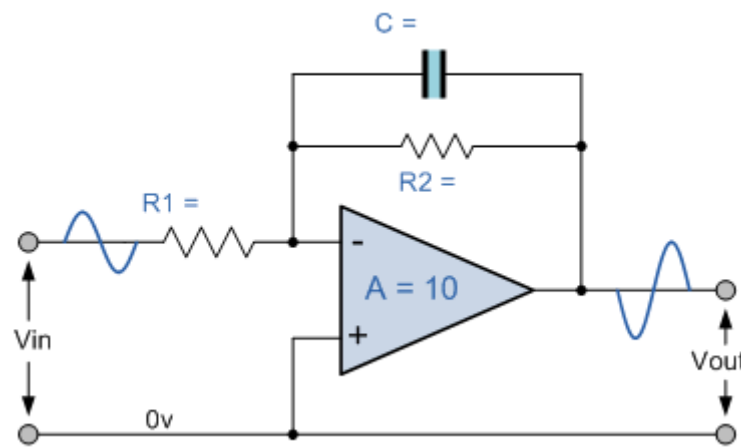
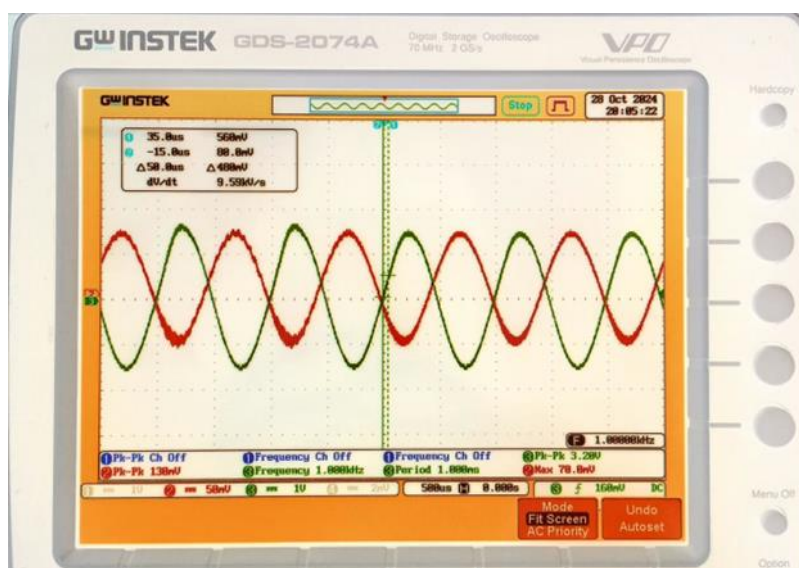


Figure 10.3: Schematic for Active low Pass filter

1. Annotate the circuit given in Figure 10.3. Calculate values for $R1$, $R2$ and $C1$ to set gain = 10 V/V and cut-off frequency = 1KHz. **Note that**, For the sake of simplicity, we will not take into account the capacitance when calculating gain.
2. Predict the output voltage of the designed filter, if the input supplied is $0.5\text{ Vpk} - \text{pk}$.
 $V_{out} = \underline{\underline{5V}}$
3. Test the designed circuit, using Function generator, use $V_{in} = 0.5\text{ Vpk} - \text{pk}$ sine function of frequency 1KHz. Connect the output of the function generator to the input of the circuit (V_{in}) and the oscilloscope probe to the output of the circuit (V_{out}). Adjust the oscilloscope settings to display the input and output waveforms on the same screen. Sketch input and output waveform on the same graph with proper labelling of the axes.

Space for waveforms:





4. Now, Set the function generator to output a 10 Hz sine wave with an amplitude of 0.5 V_{pk} – pk and fill Table 10.1.
5. Repeat the same steps for the frequencies mentioned in Table 10.1.
6. Measure the actual resistance and capacitance of the used components.

$$R2 = \underline{3.3K}$$

$$C = \underline{47nF}$$

7. Calculate the cutoff frequency of the filter using measured values, $f_c = 1 / (2\pi R_2 C)$.
 $f_c = \underline{1026}$
8. Verify the cutoff frequency by measuring the amplitude of the output waveform at frequencies both below and above the calculated cutoff frequency. Also, mention your cut-off frequency in the empty space between 1k and 1.5k and take the readings.
9. Plot the Bode plot of the filter showing cut-off frequency, pass band, and the 17 dB point. Using the semi-log graph provided with proper axes labelling and f_c .

Table 10.1: Data Table for Low Pass Filter

Frequency	10	100	500	1k	1026	1.5K	5K	10k	50k	100K	500K	1M
Vin Max (V)	544m	530m	530m	530m	530m	535m	534m	564m	544m	568m	544m	533m
Vout Max (V)	4.56	4.40	4.08	3.28	3.4	2.52	980m	552m	124m	108m	32m	50m
$Af = \frac{V_{out} Max}{V_{in} Max}$	8.38	8.3	7.7	6.19	6.42	4.71	1.83	1.034	0.232	0.202	0.06	0.094
gain(dB) = $20\log(Af)$	42.52	42.33	40.82	36.17	37.19	28.56	12.09	0.669	-29.22	-31.99	-56.27	-47.29

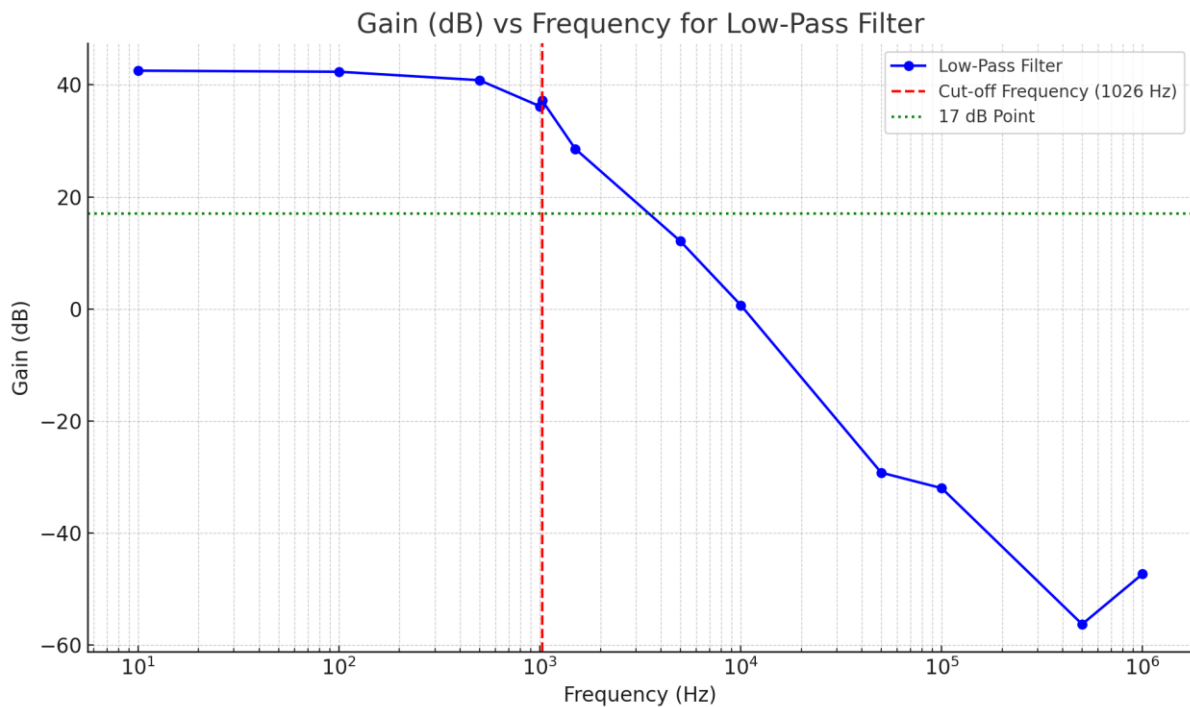


Figure 10.4: Gain(dB) Vs Frequency Graph for Low Pass Filter

10.4 Active High Pass Filter

The principle of operation of an active high pass filter and frequency response is exactly the same as that for the passive high pass filter, just like the active low pass filter, in active high pass filter we are going to use an op-amp. The active high pass filter circuit is shown in figure below.

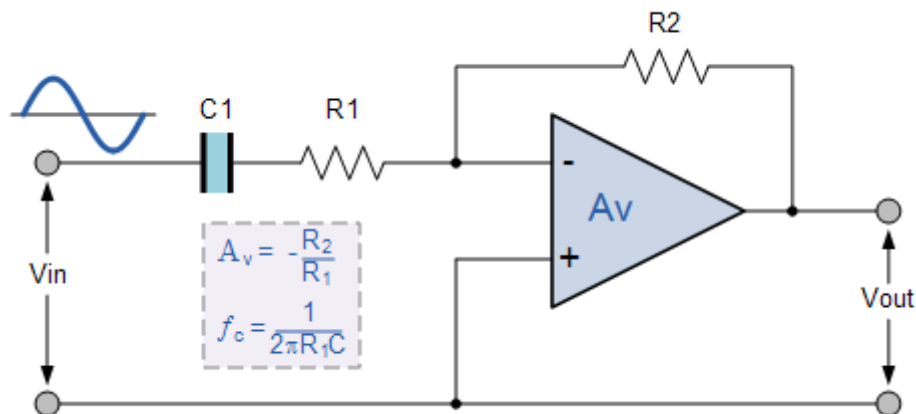


Figure 10.5: Active High Pass Filter

Technically, there is no such thing as an active high pass filter. Unlike Passive High Pass Filters which have an “infinite” frequency response, the maximum pass band frequency response of an active high pass filter is limited by the open-loop characteristics or bandwidth of the operational amplifier being used, making them appear as if they are band pass filters with a high frequency cut-off determined by the selection of op-amp and gain.

10.4.1 Cut-off Frequency

The cut off frequency for high pass RC circuits is given as:

$$f_c = \frac{1}{2\pi R_1 C_1}$$

10.4.2 Gain of Active High Pass Filter

The circuit in Figure 10.5 illustrates active filter circuitry with inverting configuration. Where resistors R_2 and R_1 set the amplifier gain² given as:

$$DC \text{ Gain } (A_F) = -\frac{R_2}{R_1}$$

If we take the effect of capacitor into account the gain to First-order high pass filter is given by:

$$\text{Voltage Gain } (A_V) = \frac{V_{out}}{V_{in}} = \frac{A_F \left(\frac{f}{f_c} \right)}{\sqrt{1 + \left(\frac{f}{f_c} \right)^2}}$$

where:

f = the frequency of the input signal in Hertz, (Hz)

f_c = the cut-off frequency in Hertz, (Hz)

Thus, the operation of a low pass active filter can be verified from the frequency gain equation above as:

1. At very low frequencies, $f < f_c$, $\frac{V_{out}}{V_{in}} < A_F$
2. At the cut-off frequency, $f = f_c$, $\frac{V_{out}}{V_{in}} \cong \frac{A_F}{\sqrt{2}}$
3. At very high frequencies, $f > f_c$, $\frac{V_{out}}{V_{in}} \cong A_F$

10.4.3 Frequency Response of an Active High Pass Filter

The Bode Plot shows the Frequency Response of the filter to be nearly flat for high frequencies and all of the input signal is passed directly to the output, resulting in a gain value A_v . However, below the Cut-off Frequency point (f_c), there is attenuation.

² For the sake of simplicity, the effect of capacitor has not been taken into account.

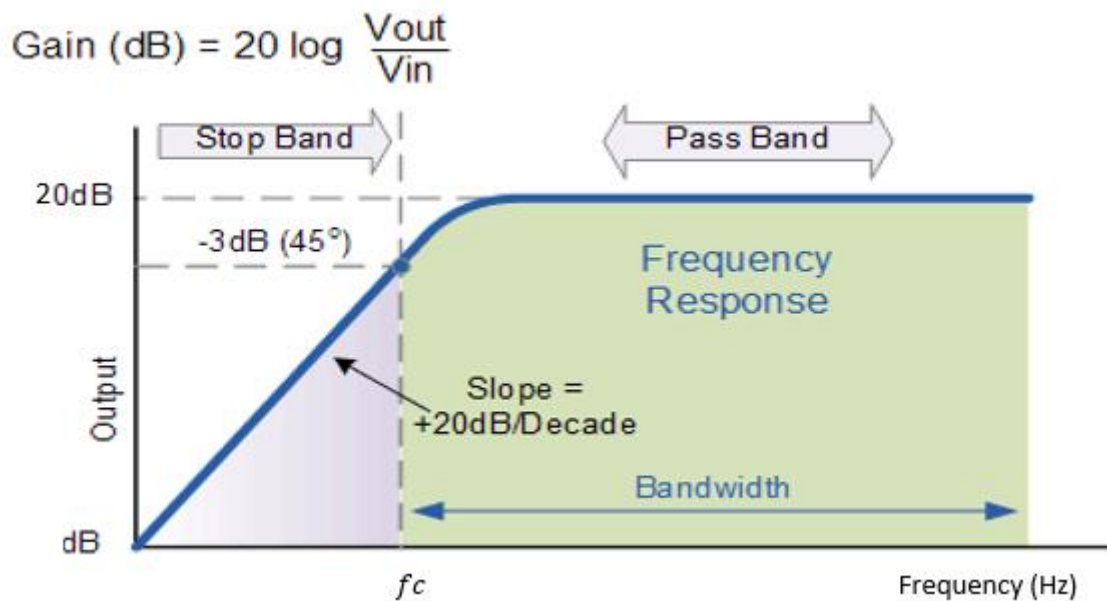


Figure 10.6: Frequency Response of an active High Pass filter

Task 2: To design passive High Pass Filter and verify with hardware implementation

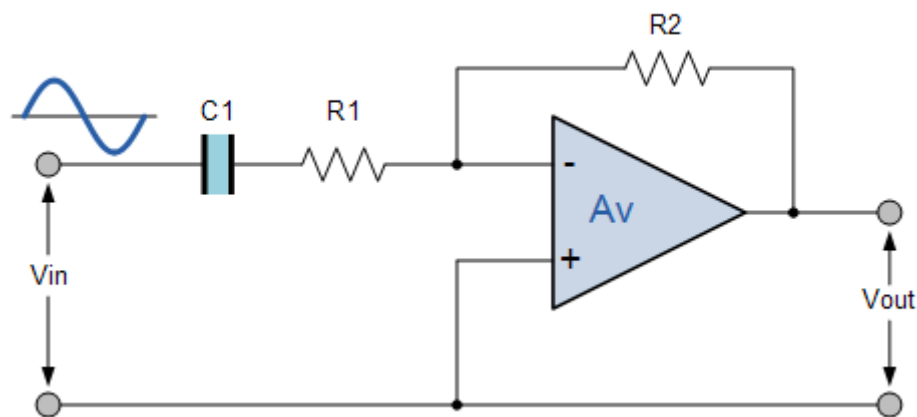
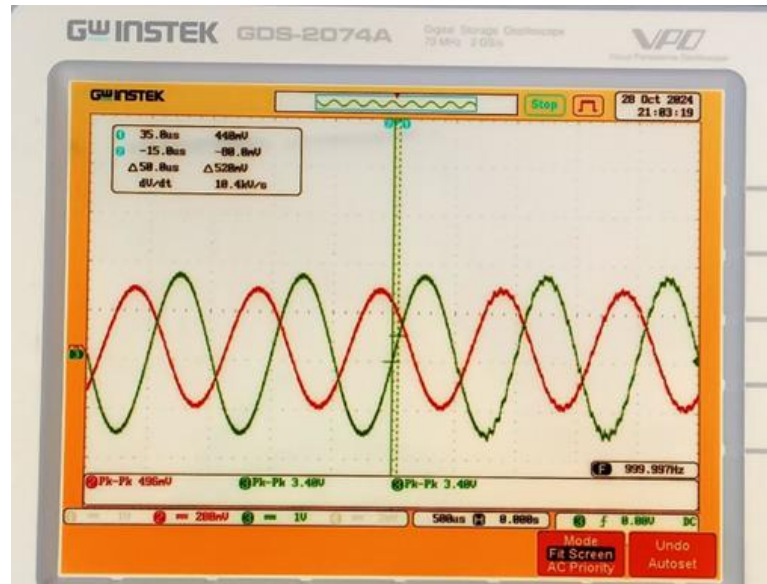


Figure 10.7: Schematic for active High Pass Filter

1. Annotate the circuit given in Figure 10.7. Calculate values for $R1$, $R2$ and $C1$ to set gain = 10 V/V and cut-off frequency = 1KHz. **Note that**, For the sake of simplicity, we will not take into account the capacitance when calculating gain.
2. Predict the output voltage of the designed filter, if the input supplied is 0.5 Vpk – pk.
 $V_{out} = \underline{\hspace{1cm}} 5V \underline{\hspace{1cm}}$
3. Test the designed circuit, using Function generator, use $V_{in} = 0.5 V_{pk} - pk$ sine function of frequency 1KHz. Connect the output of the function generator to the input of the circuit (V_{in}) and the oscilloscope probe to the output of the circuit (V_{out}). Adjust the oscilloscope settings to display

the input and output waveforms on the same screen. Sketch input and output waveform on the same graph/plot/box with proper labelling of the axes.

Space for waveforms:



4. Now, Set the function generator to output a 10 Hz sine wave with an amplitude of $0.5 V_{pk-pk}$ and fill Table 10.2.
5. Repeat the same steps for the frequencies mentioned in Table 10.2.
6. Measure the actual resistance and capacitance of the used components.
 - a. $R1 = 3.3K$ _____
 - b. $C1 = 47nF$ _____
7. Calculate the cutoff frequency of the filter using measured values, $f_c = 1 / (2\pi R_1 C_1)$.
 - a. $f_c =$ 1026 _____
8. Verify the cutoff frequency by measuring the amplitude of the output waveform at frequencies both below and above the calculated cutoff frequency. Also, mention your cut-off frequency in the empty space between 1k and 1.5k and take the readings.
9. Plot the Bode plot of the filter showing cut-off frequency, pass band, and the 17 dB point. Using the semi-log graph provided with proper axes labelling and f_c .

Table 10.2: Data Table for High Pass Filter

Frequency (Hz)	10	100	500	1k	1026	1.5K	5K	10k	50k	100K	500K	1M
Vin Max (V)	536 m	536 m	536 m	530 m	520m	560m	536 m	504 m	504 m	510m	510m	512 m
Vout Max (V)	224 m	640 m	2.20	3.36	3.6	4.4	4.96	4.96	4.08	2.72	368m	112 m
$A = \frac{V_{out\ Max}}{V_{in\ Max}}$	0.418	1.194	4.1	6.27	6.72	8.21	9.25	9.841	8.095	4.45	0.722	0.219
$gain(dB) = 20\log(A)$	-17.45	3.55	28.2	36.7	38.1	42.11	44.5	45.73	41.82	29.86	-6.51	-30.4

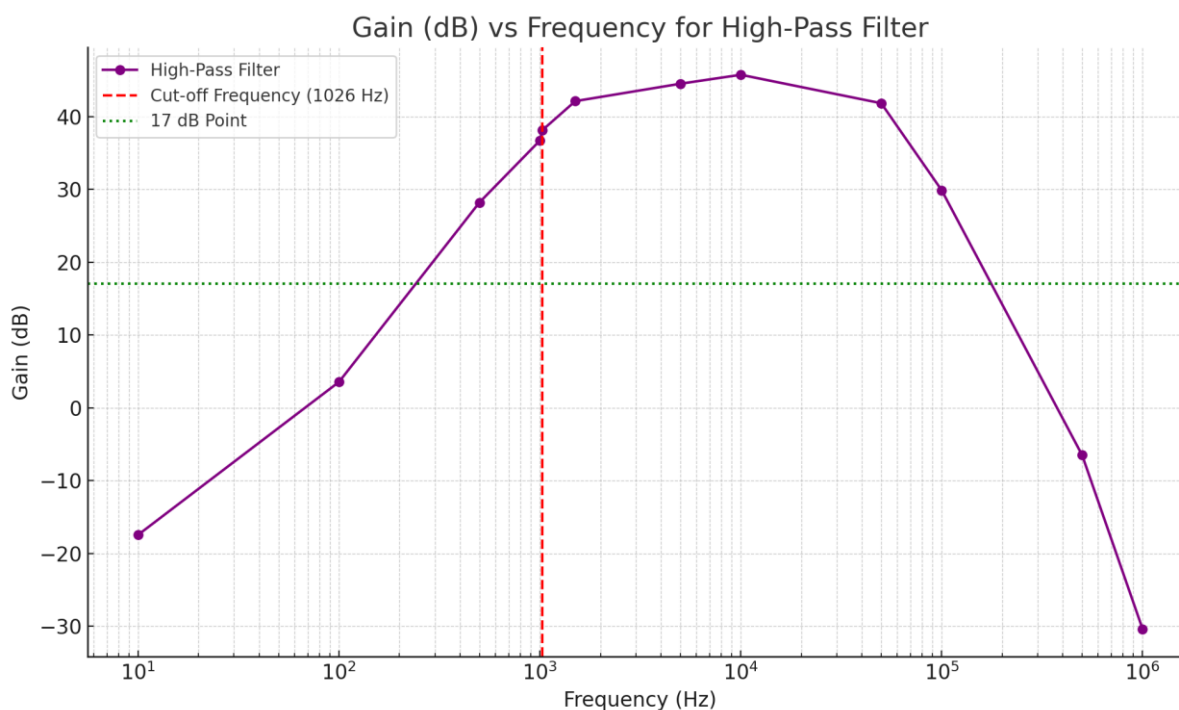


Figure10.8: Gain(dB) Vs Frequency Graph for High Pass filter



Assessment Rubric

Lab 10

Active Filters

Name:	Student ID:
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Points Distribution

Task No.	LR 7 Viva	LR 4 Data Collection	LR5 Results/Plots	LR 6 Calculation	LR10 Design
Task 1	/10	/15	/10	/5	/10
Task 2		/15	/10	/5	/10
Total	/10	/30	/20	/10	/20
CLO Mapped	CLO 2	CLO 2	CLO 2	CLO 2	CLO 2

Affective Domain Rubric		Points	CLO Mapped
AR 4	Report Submission	/10	CLO 4

CLO	Total Points	Points Obtained
2	90	
4	10	
Total	100	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.



Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3: Good Points 3	Level 4: Exemplary Points 4
LR1	Circuit Layout	Connections between circuit components are mostly wrong. Circuit layout is cluttered. Needs guidance to make correct equipment/ component connections.	Few of the connections made between circuit components are wrong. Circuit layout is not neat and clean as per standards.	Circuit layout and connections are correct but not all connections are neat and clean as per standards.	Neat, clean and correct connections of all the circuit components/ equipment are made as per standard circuit diagram.
AR4	*Report Submission	Not accepted after 1 week.	Late submission after 2 days and within 1 week.	Late submission after the lab timing and within 2 days of the due date.	Timely submission of the report and in the lab time.
LR7	Viva	Response shows a complete lack of understanding of the assigned / completed task	Response shows shallow understanding of the assigned task	Response shows substantial understanding of the assigned task	Response shows complete understanding of the completed task. The student is able to explain all the related concepts.
LR4	Data Collection	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in using symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and task completed in due time.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR6	Calculations	Formulae used and/or calculations are incorrect. Units are not mentioned with results.	Formulae used are correct but there are few mistakes in calculation which lead to incorrect results.	Formulae used and end results are correct. Complete working steps are not shown. Proper units are not mentioned with results.	Formulae used, end results and all calculations are correct with all the intermediate steps clearly shown. Units are mentioned with results.
LR9	Report	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.
LR10	Analysis	Results are not interpreted or are analyzed incorrectly.	Analysis of the obtained results is incomplete. Conclusions are not drawn.	Results are analyzed and concluded but are not well explained and justification is not provided with the help of reasoning.	Results are well analyzed supported with reasons. Good comparison is made between the theoretical and experimental results with correct and useful conclusion.
LR11	Design	Proposed design is unsubstantiated or does not satisfy most of given constraints. The breakdown of system into features is incomplete and still at lower resolution.	Justification for proposed design is weak, and it only satisfies some constraints. The breakdown of system is missing some features and resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.